Loose lips sink ships: Mitigating Length Bias in Reinforcement Learning from Human Feedback

Wei Shen[1*], **Rui Zheng**[1*], **Wenyu Zhan**[1], **Jun Zhao**[1], **Shihan Dou**[1], **Tao Gui**[3†], **Qi Zhang**[1], **Xuanjing Huang**[1,2†]

[1] School of Computer Science, Fudan University

[2] International Human Phenome Institutes (Shanghai), Shanghai, China

[3] Institute of Modern Languages and Linguistics, Fudan University

wshen21@m.fudan.edu.cn,
{ruizheng20,tgui,xjhuang}@fudan.edu.cn

Abstract

Reinforcement learning from human feedback serves as a crucial bridge, aligning large language models with human and societal values. This alignment requires a vast corpus of human feedback to learn a reward model, which is subsequently used to finetune language models. However, we have identified that the reward model often finds shortcuts to bypass its intended objectives, misleadingly assuming that humans prefer longer responses. The emergence of length bias often induces the model to favor longer outputs, yet it doesn't equate to an increase in helpful information within these outputs. In this paper, we propose an innovative solution, applying the Product-of-Experts (PoE) technique to separate reward modeling from the influence of sequence length. In our framework, the main expert concentrates on understanding human intents, while the biased expert targets the identification and capture of length bias. To further enhance the learning of bias, we introduce perturbations into the bias-focused expert, disrupting the flow of semantic information. Experimental results validate the effectiveness of our approach, indicating that language model performance is improved, irrespective of sequence length.

[Human]: What does the phrase "a stitch in time saves nine" mean?
[Assistant]: It's a common phrase in English that means " an early solution to a problem saves a lot of effort in the long run ".
[Human]: Can you apply this saying **in a sentence**?

SFT model
[Assistant]: Sure, here's an example sentence: "My brother always says that a stitch in time saves nine, so he never procrastinates on fixing things."

Vanilla RM + PPO
[Assistant]: Sure, here's an example: "I always make sure to fix any small issues in my car straight away rather than waiting for them to become bigger problems. A stitch in time saves nine, they say!"

Ours + PPO
[Assistant]: Yes, here's an example of how you can apply this saying in a sentence: "Fixing that leak in the pipe right away is a stitch in time saves nine. "

Figure 1: Comparison of model outputs using SFT, Vanilla RM with PPO, and Our Method with PPO. Our study demonstrates that the current reward model tends to incentivize the model to generate longer responses, disregarding the true human intent. This phenomenon is highlighted in the example shown, where we observe a decline in model performance.

1 Introduction

In recent years, the field of natural language processing has witnessed remarkable advancements with the emergence of powerful models like InstructGPT (Ouyang et al., 2022), GPT-4 (OpenAI, 2023), Claude (Bai et al., 2022b), and others. These models have displayed remarkable proficiency in understanding human queries and providing helpful responses. Their success can be attributed to a two-step learning process that involves Supervised Fine-Tuning (SFT) followed by the utilization of Reinforcement Learning from Human Feedback (RLHF) techniques (Ouyang et al., 2022; Bai et al., 2022a). This combination enables these models to not only learn how to follow human instructions but also better understand human intent and align with human and societal values. Undoubtedly, RLHF plays a pivotal role in the success of these models.

One of the key components of RLHF is reward modeling (RM), which involves learning a reward function from human preferences or demonstrations. This allows an RL agent to optimize its behavior based on the feedback received from reward model (Ziegler et al., 2019b). However, the process is not without challenges. Human preference data can be noisy and subject to inconsistencies among different annotators, leading to suboptimal results (Bai et al., 2022a).

*Equal contribution.
†Corresponding author.

For example, reward gaming, a well-known and pervasive issue, refers to the phenomenon where trained models exhibit undesirable patterns and generate low-quality outputs while still receiving high rewards (Skalse et al., 2022; Pan et al., 2022). These complexities emphasize the need for careful consideration and robust methods to ensure reliable and meaningful reward functions in RLHF.

As shown in Figure 1, a similar reward gaming issue arises in NLP reward modeling. We have observed that the reward model tends to reply on simple patterns, such as sentence length, to differentiate between good and bad responses. Typically, the reward model assumes that longer responses are better, which hinders its ability to learn the true human intent and preference. Addressing this problem is crucial for improving the effectiveness of NLP reward modeling and capturing the true nuances of human language.

In this paper, we propose a Product-of-Experts (PoE)-based method (Hinton, 2002) that consists of two expert models to decouple human intent and response length during the reward modeling phase. The first expert operates similarly to a standard reward model and focuses on learning the true human intent behind responses. The second expert, referred to as the bias-only expert, is designed to learn simple patterns, specifically the length of responses. It employs a smaller model capacity and a larger learning rate to capture coarse-grained information of inputs. Additionally, stochastic perturbations are introduced into the inputs of the bias-only expert, intentionally disrupting the semantic information present in the input. To summarize, the main contributions of our work are followings:

- We identify that reward modeling in NLP tends to rely on length bias, hindering the models from accurately learning true human intent and even leading to model degradation.
- We propose a simple and efficient solution leveraging PoE technique. Our method effectively decouples length bias from human intent, enabling models to better capture and understand human preferences.
- We validate the effectiveness of our proposed method. The results show that our approach enhances the learning of human intent by avoiding the generation of meaningless and overly verbose outputs.

2 Related Work

Reinforcement Learning from Human Feedback. Using human preference feedback is a popular way for realizing AI alignment (Leike et al., 2018). Preferences are often provide numerical value or demonstrations (WirthChristian et al., 2017) without requiring expert proficiency or fine-grained feedback . Alignment bring a potent capability to state-of–the-art generative foundation models, like InstructGPT (Ouyang et al., 2022), Sparrow (Glaese et al., 2022), Claude (Bai et al., 2022b), which means this method is of great success in the paradigm of learning with human feedback. Some prior work have explored using human feedback to improve various tasks, such as summarization (Stiennon et al., 2022; Ziegler et al., 2019b), dialogue (Bai et al., 2022a), translation (Bahdanau et al., 2016), event generation (Zhou and Xu, 2020), semantic parsing (Lawrence and Riezler, 2019) and instruction following (Ouyang et al., 2022; Bai et al., 2022a). These work can be categoried as supervised fine-tuing or reward modeling on well-constructed human annotations information, the latter is also known as a vital phase in RL from human feedback (Christiano et al., 2023; MacGlashan et al., 2017). Our work falls within the realm of RLHF and aims to awaken LM with both harmless and helpful abilities.

Reward Hacking. Goodhart's Law[1] (Strathern, 1997) can be formulated a tough challenge in numerous fields. A few approaches have been proposed for reducing overoptimization in general reinforcement learning (Everitt et al., 2017), as well as in reward models (Gleave and Irving, 2022). The overoptimization problem of reward model can generally regard as a special case of reward gaming, also known as reward hacking (Skalse et al., 2022). In addition, Pan et al. (2022) proposed to systematically analyze reward misspecification in RL by creating a set of domains where the agent optimizes a hand-engineered proxy reward function. In this study, we consider the length of human preferences as a confounding factor that hinders the reward model from accurately assessing the quality of model responses based on true human intent.

Products-of-Experts. Products-of-Experts (PoE) (Hinton, 2002) has been proposed as an alternative to mixture model to compensate for their poor

[1]Goodhart's law is an adage often stated as, *When a measure becomes a target, it ceases to be a good measure*

efficiency in high dimensional space. This technique is often based on the principle of *wisdom of crowds*, which suggests that aggregating multiple models can lead to better performance than relying on a single model. Clark et al. (2019) firstly use PoE to build a paradigm that train a debiased model ensemble with a bias-only model. The goal is to encourage the debiased model to utilize orthogonal information with information from the bias-only model. Typically, this kind of method (Clark et al., 2019; He et al., 2019) usually contains two stages. In this work, we adopt the end-to-end manner like (Karimi Mahabadi et al., 2020) which jointly learn the bias-only model and the debiased main model simultaneously, therefore, our reward model can take advantage of PoE to use a weak learner to capture the length shortcuts without any prior information about the length of sentences, and the main model can purely attain the correct knowledge that is suitable for human preference.

3 Preliminary

For the purpose of implement RLHF, we follow the pipeline in Ziegler et al. (2019a). It is usually made up of three phrases: 1) supervised fine-tuning, 2) reward modeling, 3) reinforcement-learning optimization, our attention is directed towards the last two phases.

SFT: It begins with a generic pre-trained LM, which is fine-tuned using supervised learning on a high-quality instruction dataset. This allows the model to follow various instructions, perform dialogue and dialogue. As a result, we obtain an LM π^{SFT} during this phase.

Reward Modeling: According to the Bradley-Terry model (Bradley and Terry, 1952), a reward function is hard to describe and needs to be learned from preferences among trajectories. The reward model r_θ is trained on human preference dataset to predict which response $y \in \{0, 1\}$ is better as judged by human, given a query x. If the response preferred by human is y_i, the RM loss can be expressed as:

$$-\mathbb{E}_{(x,y)\sim\mathcal{D}}[\log(\sigma(r_\theta(x, y_i) - r_\theta(x, y_{1-i})))], \quad (1)$$

where $r_\theta(x, y)$ is the scalar output of the reward model for query x and response y with parameters θ, and $\mathcal{D}$ is the human preference dataset.

RL Optimization: Then we fine-tune the SFT model on our environment using PPO (Schulman et al., 2017). The language model is provided with feedback through the launch of a learned reward function. To this end, we maximize the following objective function in RL training:

$$\mathbb{E}_{(x,y)\sim\mathcal{D}}\left[r_\theta(x, y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right],$$

β is a parameter to control the deviation from the SFT model π^{SFT}. The language model policy π_ϕ is initialized to π^{SFT}. Importantly, the last per-token KL-penalty term is used to prevent the model from deviating too far from the deviating exceeding the appropriate scope from the distribution on which the reward model is accurate, as well as maintaining the generation diversity and preventing mode-collapse to single high-reward answers (Ziegler et al., 2019b).

4 Length Bias in Reward Model

In this section, we present the phenomenon of length bias in reward models and utilize causal analysis to examine the underlying reasons for this occurrence. We shed light on why the reward model exhibits a preference for longer sentences and delve into the causal factors contributing to this bias.

4.1 Length Bias Phenomenon

We present a Figure 2 depicting the scores and lengths of 4000 SFT model output results, which were evaluated using the vanilla reward model trained on the helpful and harmless (HH) dataset (Bai et al., 2022b). It is evident that there is a strong correlation between the reward scores and lengths. When the model generates longer sequences, the reward model tends to assign higher scores. This correlation contradicts human intent since the helpfulness and harmlessness of the output should not be solely determined by its length. More figures can be seen in the Appendix 8.

In addition, the length bias locates in PPO as well, we additionally investigate it on TL;DR (Stiennon et al., 2022) in the Appendix A.4

4.2 Confounding Factor

As Figure 3 depicted, we formulate the problem as a causal structure (Zeng et al., 2023; Tien et al., 2023) of preference-based reward modeling. Pairwise preference information are conveyed based on an observed reward function r. Features (x, y) are causal factors can affect r and another nuisance features z regarded as a potential confounder factor that have impact on r. Note that